

MoleGraph v4.3 (EN version)

<https://github.com/e-Mole/MoleGraph>

Supported Sensors Overview (sensors.json)

Configuration: <http://www.tfsoft.cz/molegraph/json>

Editor: <http://www.tfsoft.cz/molegraph/json/dev>

ID	Sensor Name	Measured Quantities (Units)	Description
101	Generic (AD 10bit)	RAW, Voltage (V), Trigger, Trigger count	Arduino NANO AD 10-bit converter as RAW value [0-1023], input voltage [0-5 V] or trigger [0/1].
1	Thermometer (DS18B20)	Temperature (°C)	Measuring range -55..+125 °C. Output Resolution: 9-bit to 12-bit (programmable), accuracy: ±0.5 °C. Conversion time: 750 ms at 12-bit.
5	IR Thermometer (MLX90614)	Temperature (°C), Temperature (°C)	Measuring range: -40..+125 °C for sensor temperature and -70..+380 °C for object temperature. Resolution 0.02 °C, accuracy ±0.5 °C. Maximum sampling rate 1000 Hz.
34	Thermometer (K-Type, MAX6675)	Temperature (°C)	Measuring range -200..+1200 °C. Resolution 0.25 °C, accuracy around ±2 °C. Maximum sampling rate 4 Hz. Error state (no thermocouple) send value -500.
2	Pressure + Thermometer + Hygrometer (BME280)	Pressure (Pa), Temperature (°C), Humidity (%), Altitude (m), Dewpoint (°C)	Measuring range pressure 300..1100 hPa, temperature -40..85 °C, rel. humidity 0..100 %. Resolution pressure 0.18 Pa, temperature 0,01 °C, humidity 0,008%, accuracy pressure ±1.5 hPa, humidity ±0,5 °C, humidity ±3 %. Maximum sampling rate 1000 Hz. Operating Temperature: -40 to 85 °C.
6	Pressure sensor (MPX5700DP)	Pressure (Pa)	Measuring range: 0..750 kPa. Resolution 0.7 kPa (10-bit), accuracy ±2.5 %. Maximum sampling rate 1000 Hz. Operating Temperature: -40 to 125 °C.
14	pH sensor	pH, Voltage (V), RAW, Trigger	Measuring range 0..14 pH. Resolution 0.02 pH (10-bit), accuracy: ±0.1. Analog output - calibration is needed!
37	ORP sensor	RAW, Voltage (mV), Trigger	Measuring range -1999...1999 mV. Resolution 5 mV (10-bit), accuracy: ±10 mV. Analog output - calibration is needed! Value calculated from output voltage [0-4 V], RAW value [0-819], trigger [0/1].
15	Conductivity sensor	Conductivity (µS/cm), Salinity (ppm), RAW, Voltage (V), Trigger	Conductivity/TDS - Total Dissolved Solids. Measuring range: conductivity 0...1650 µS/cm, salinity (TDS) 0...1000 ppm. Resolution 2 µS/cm (1 ppm) (10-bit), accuracy: ±5 %. Analog output - calibration is needed!
19	CO2 gas sensor (MH-Z16)	Concentration (ppm), Concentration (%)	Measuring range 0..50 000 ppm. Resolution 0.05 ppm, accuracy around ±50 ppm. Maximum sampling rate 0.5 Hz.
20	O2 gas sensor (ME2-O2)	Concentration (ppm), Concentration (%)	In development...
36	Smoke/Combustible Gas (MQ-2)	Concentration (ppm), Concentration (%), Voltage (V), RAW, Trigger	Measuring range 0..10 000 ppm. Resolution 10 ppm (10-bit), accuracy around ±15 %. Analog output - calibration is needed!
25	Alcohol gas sensor (MQ-3)	Concentration (ppm), Concentration (%), Voltage (V), RAW, Trigger	Measuring range 25..500 ppm. Resolution 2.5 ppm (10-bit), accuracy around ±10 ppm. Analog output - calibration is needed!
38	Turbidity sensor (TS-300B)	Turbidity (NTU), Voltage (V), Trigger	Measuring range 0..1000 NTU. Resolution 5 NTU (10-bit), accuracy around ±30 NTU. Analog output - calibration is needed! Data from AD 10-bit converter as NTU value [0-1000], output voltage [0-4 V] or trigger [0/1].
42	Radioactivity	Radioactivity (cpm)	In development...

	(RadiationD v1.1)		
22	Heart rate sensor (simple)	RAW, Frequency (BPM), Trigger	Sensor detects heart pulse and calculate BPM value. The pressure with your finger on the sensor must not be too big or too small :-).
10	ECG sensor (AD8232)	Voltage (V), Frequency (BPM), Trigger	Sensor detects heart pulse signal and other biosignals. More typical electrodes placement is possible, one of the options: RA -> Red -> Right Arm, LA -> Yellow -> Left Arm, RL -> Green -> Right Leg.
41	Spirometer (MPXV7002DP)	RAW, Voltage (V), Trigger	In development...
28	DC Voltage sensor 0-25 V (voltage divider)	Voltage (V), RAW	Measuring range 0..25 V. Resolution 25 mV (10-bit), accuracy around ± 25 mV.
29	AC/DC Current sensor 0-5 A (ACS712)	Current (A), RAW	Measuring range 0.. ± 5 A. Resolution 10 mA (10-bit), accuracy around ± 1.5 %.
30	AC/DC Current sensor 0-30 A (ACS712)	Current (A), RAW	Measuring range 0.. ± 30 A. Resolution 60 mA (10-bit), accuracy around ± 1.5 %.
16	Voltage + Current sensor (HX711)	Voltage (V), Current (A)	In development...
18	Light sensor (photoresistor)	Illuminance (RAW), Voltage (V), Trigger	Measuring range 0..5 V (10-bit converter) – output only as RAW value 0..1023, voltage 0..5 or trigger 0/1.
32	Light sensor (BH1750 luxmeter)	Illuminance (lux)	Measuring range 0..65 535 lux (16-bit converter). Resolution 1 lux, accuracy 1 lux. Measurement time is approx 120 ms (max sampling 8 Hz).
13	UV sensor (GUVA-S12SD)	UV Index	In development...
31	UV sensor (VEML6070)	UV Intensity	UV spectrum sensitivity: 320-410 nm (peak at 355 nm), UVA Sensitivity 5 $\mu\text{W}/\text{cm}^2/\text{step}$ (typical). Maximum sampling rate 8 Hz.
11	Color sensor (TCS3200)	RAW	In development...
26	Particle/Dust sensor (DSM501)	Concentration (ppm)	In development...
27	Sound sensor (microphone)	RAW, Voltage (V), Trigger	Measuring range 0..5 V (10-bit converter) – output only as RAW value 0..1023, voltage 0..5 or trigger 0/1.
23	Magnetic field sensor	RAW, Voltage (V), Trigger	Measuring range 0..5 V (10-bit converter) – output only as RAW value 0..1023, voltage 0..5 or trigger 0/1.
3	Accelerometer + Magnetometer (LSM303DLHC)	Acceleration (m/s^2), Magnetic Field Intensity (μT), Accel X (m/s^2), Accel Y (m/s^2), Accel Z (m/s^2), Mag X (μT), Mag Y (μT), Mag Z (μT), Heading ($^\circ$), Roll ($^\circ$), Pitch ($^\circ$)	Measuring range ± 16 g and ± 8.1 gauss (auto sensitivity). Resolution ± 2 mgauss, accuracy ± 60 mg.
7	Force sensor (HX711)	Force (N)	24-Bit Analog-to-Digital Converter (ADC) for Weigh Scales. Full scale differential input range $\pm 0.5(\text{AVDD}/\text{GAIN})$ V.
21	Motion/distance US sensor (SR-HC04)	Distance (m), Velocity (m/s), Acceleration (m/s^2)	Measuring range 2..400 cm, measuring angle 15° . Accuracy ± 3 mm. The Module automatically sends eight 40 kHz and detect whether there is a pulse signal back. Suggest to use over 60 ms measurement cycle, in order to prevent trigger signal to the echo signal.
105	Motion/distance laser sensor (VL53L0X)	Distance (m), Velocity (m/s), Acceleration (m/s^2)	Measuring indoor short range 1..120 cm (long range 1..220 cm, white target), laser beam 940 nm, measuring angle 25° . Accuracy ± 4 %. Typical timing budget for a range is 33 ms.
39	Photogate (810H)	Pulse duration (On) (s), Pulse duration (Off) (s), Period (Rising) (s), Period (Falling) (s), Frequency (Rising) (Hz), Frequency (Falling) (Hz), Pulse count (Rising)	Same data processing as sensor [Timer GPIO 01]. RAW value [0-1023], input voltage [0-5 V] or trigger [0/1] can be obtained after setting as [Generic (AD 10bit)] sensor.
35	Length sensor (Caliper)	Length (mm)	The measurement range depends on the caliper used. Resolution 0.01 mm. Accuracy 0.05 mm. Maximum

			sampling frequency 5 Hz.
103	Timer GPIO 01	Pulse duration (On) (s), Pulse duration (Off) (s), Period (Rising) (s), Period (Falling) (s), Frequency (Rising) (Hz), Frequency (Falling) (Hz), Pulse count (Rising)	Measuring range up to 2 kHz. Timer is set for GPIO 01 only.
40	Output/input prototype breadboard	None	Only for visual coding tests...